Automated Network Request Management

Phase 4: Project Development & Testing Phase — Flow Designer Build, Portal Verification, ACL Security & Audit Validation

1. Flow Designer Automation Build & Step-by-Step Logic

The core automation engine was constructed using ServiceNow **Flow Designer** (Workflow Studio) in the Global application scope. The flow eliminates custom script includes and legacy workflow engines, utilizing native actions to achieve end-to-end orchestration.

> **Flow Architecture:** Trigger [Service Catalog] ➔ Action 1 [Get Catalog Variables] ➔ Action 2 [Create Database Tables Record] ➔ Action 3 [Send Email Confirmation] ➔ Action 4 [Ask for Approval] ➔ Flow Logic 5 [IF Approved: Update Record] ➔ Flow Logic 7 [ELSE: Update Record as Rejected].

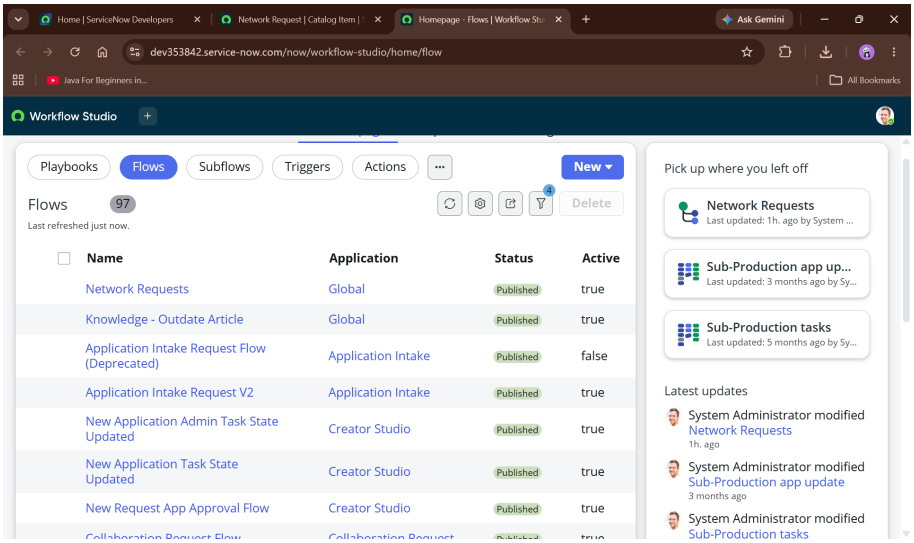


Figure 1: Workflow Studio / Flow Designer Homepage in instance dev353842

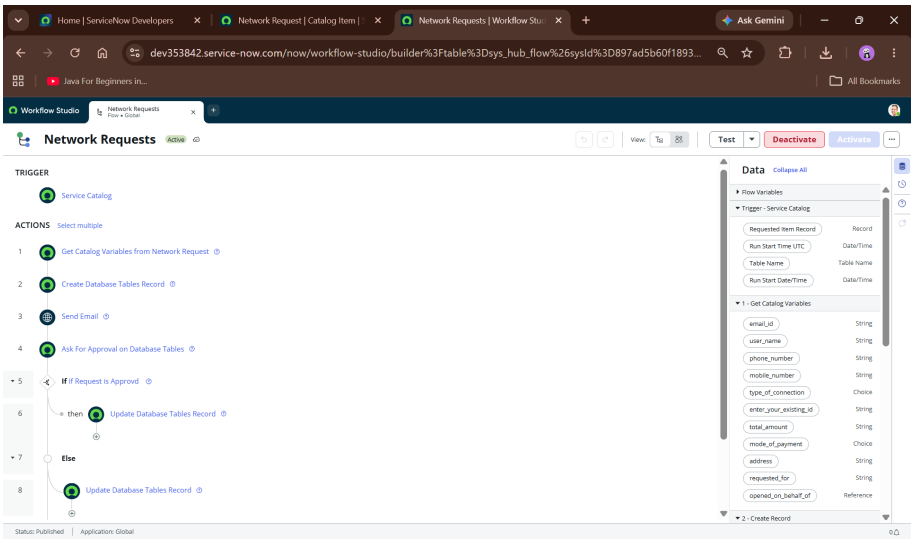


Figure 2: Flow Builder Trigger Setup — Application: Service Catalog (Fires on Catalog Item Submission)

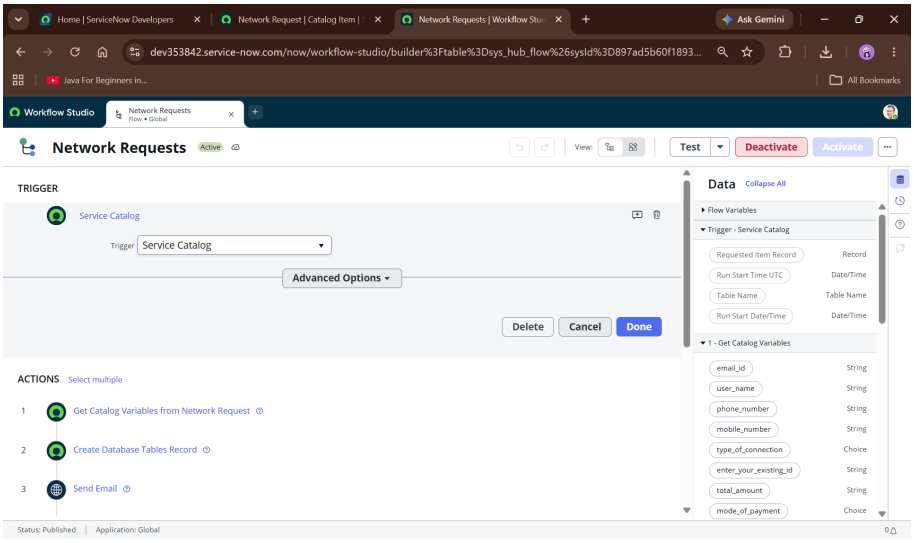


Figure 3: Action 1 — Get Catalog Variables from Trigger Requested Item for 'Network Request'

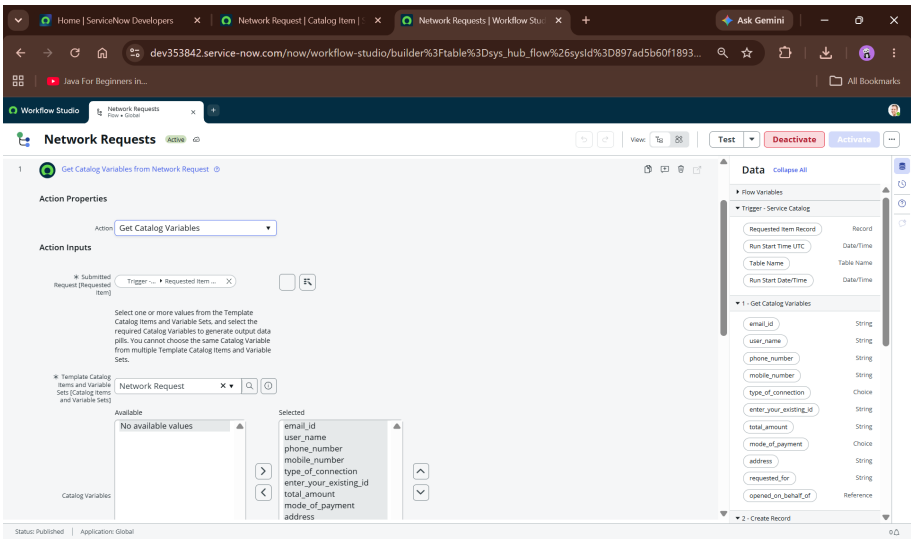


Figure 4: Action 2 — Create Record in Database Tables (u_database_tables) with Variable Data Pill Mappings

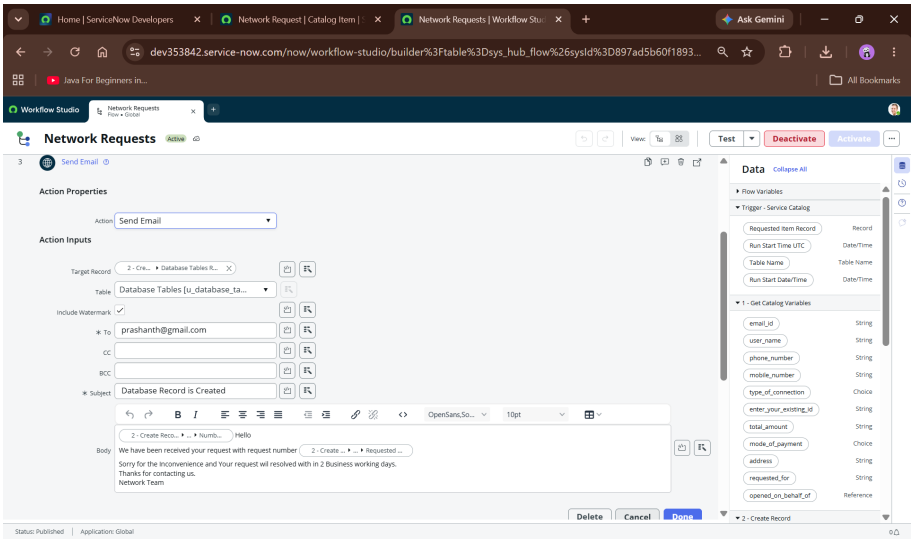


Figure 5: Action 3 — Send Email Action Dispatching Request Confirmation Notification to Requester

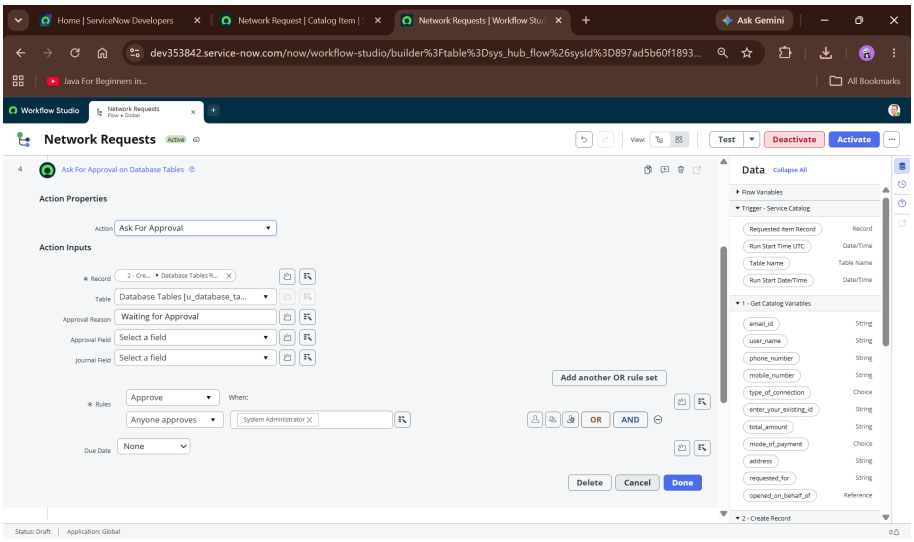


Figure 6: Action 4 — Ask For Approval on Target Database Tables Record (Anyone Approves Rule)

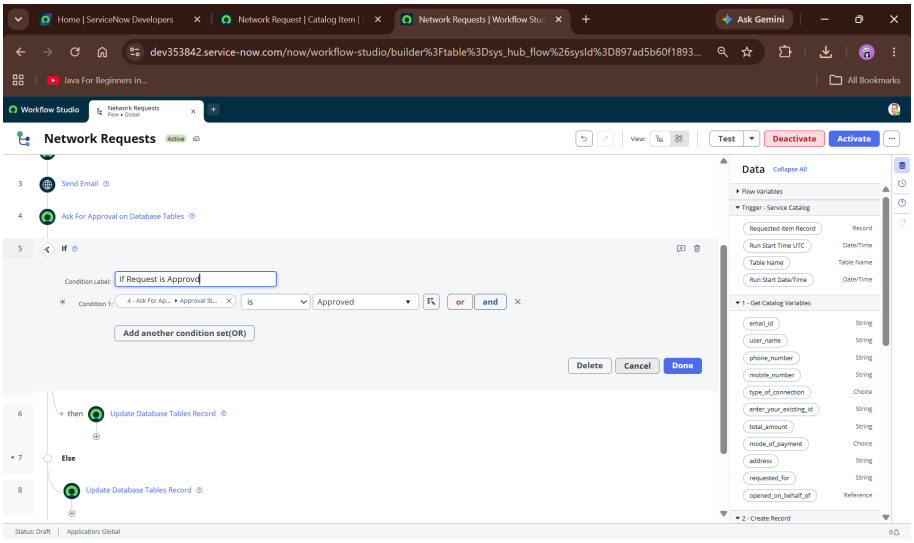


Figure 7: Flow Logic 5 — IF Condition Evaluating Ask for Approval State == 'Approved'

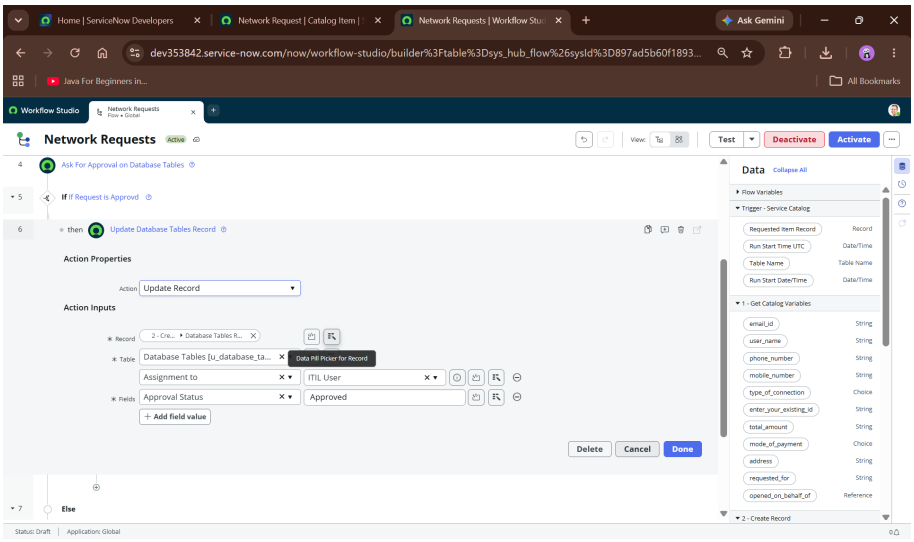


Figure 8: Action 6 — Update Database Tables Record (Setting Approval Status to 'approved')

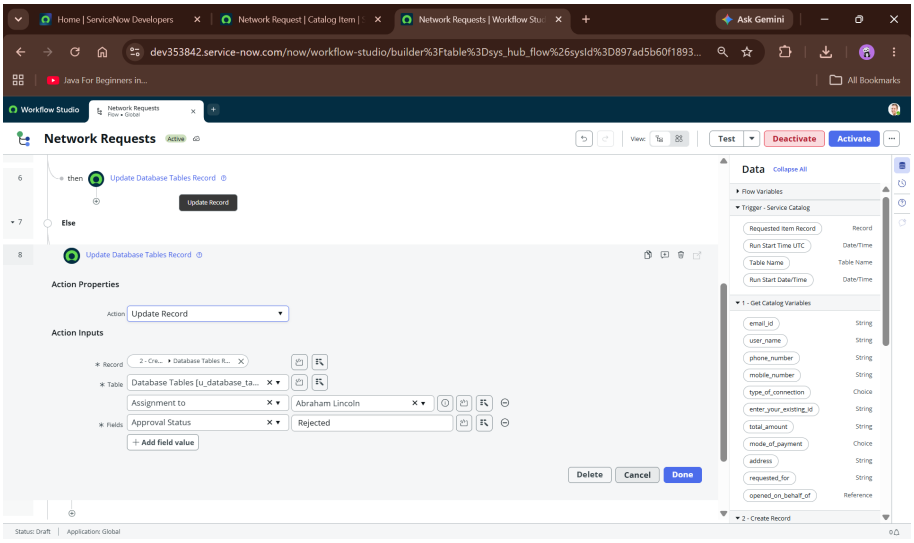


Figure 9: Flow Logic 7 — ELSE Branch Handling Rejections and Setting Status to 'rejected'

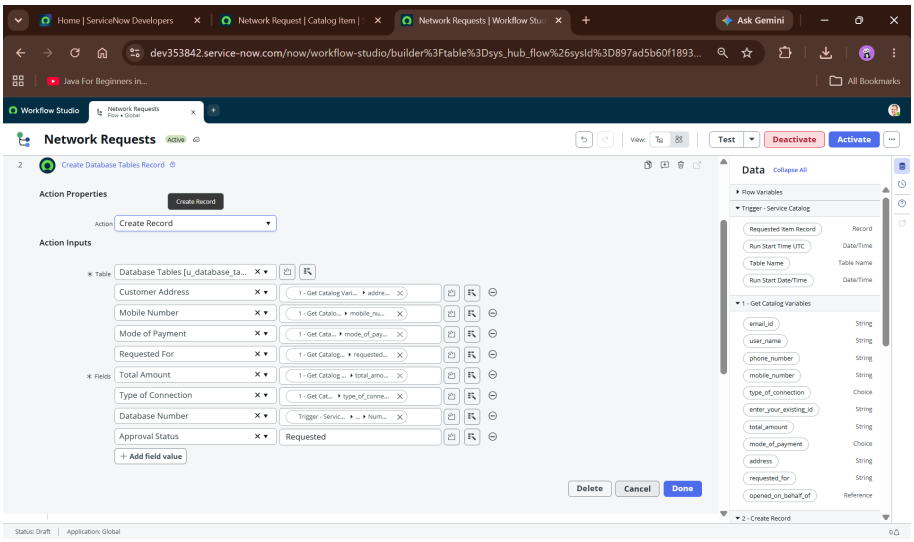


Figure 10: Complete Active Flow Designer Pipeline — Network Requests Flow Successfully Activated

2. End-to-End Service Portal User Experience & Request Submission

The complete end-user request lifecycle was verified in the modern ServiceNow Service Portal (/sp). A realistic test submission was conducted under user identity 'satya' with recipient 'anusri':

- **Access Portal:** Navigate to https://dev353842.service-now.com/sp.
- **Catalog Discovery:** Search 'network request' in the search bar and select the Network Request item.
- **Form Completion:** Populate Requested For (satya), User Name (anusri), Mobile Number (7816006060), Type of Connection (NEW), Mode of Payment (UPI), Total Amount, and Email.
- **Order Confirmation:** Review summary and click *Order Now* / *Submit*.
- **Request Generation:** System successfully generates Request number **REQ0010003** with Estimated Delivery Stage 'Request Approval'.

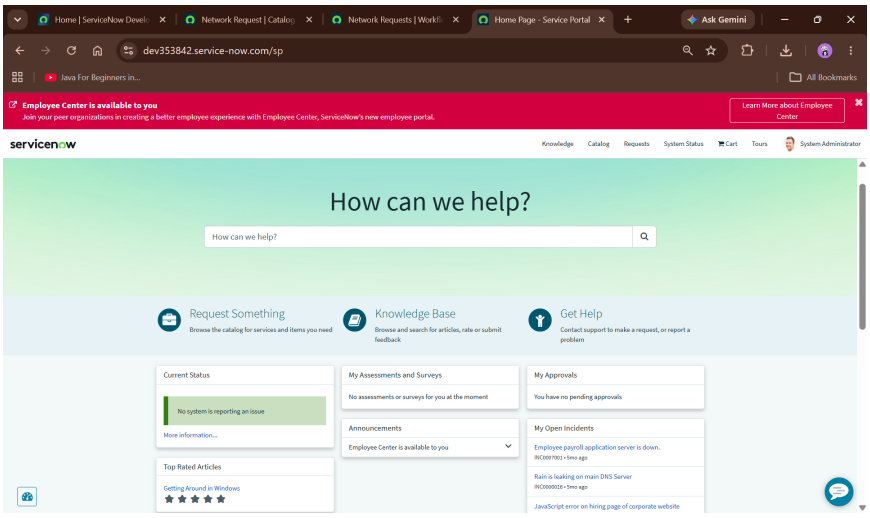


Figure 11: ServiceNow Service Portal Homepage (/sp) in Personal Developer Instance dev353842

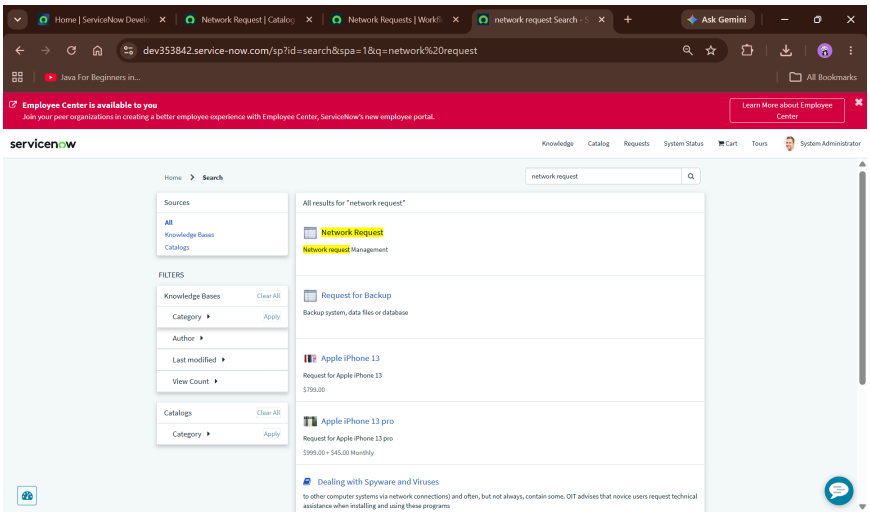


Figure 12: Service Portal Catalog Search locating 'Network Request' Item

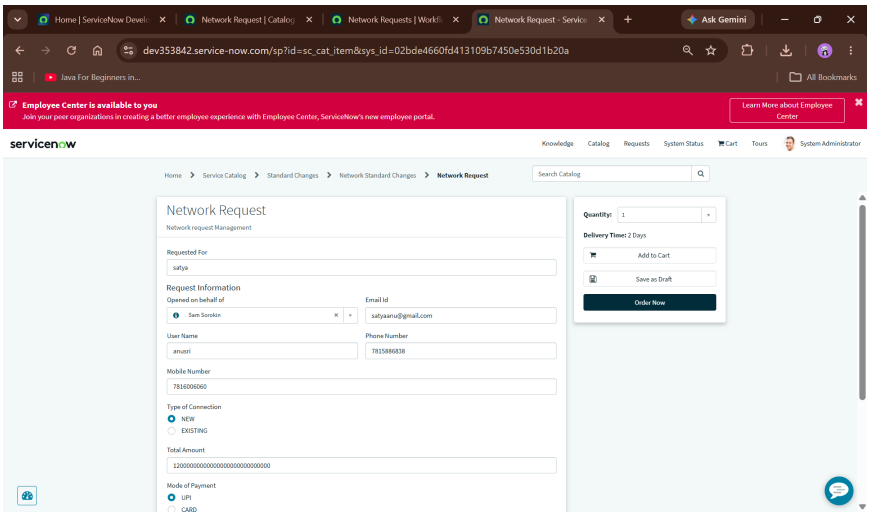


Figure 13: Service Portal Form Fill — Populating Network Request Parameters (User: satya, Payment: UPI)

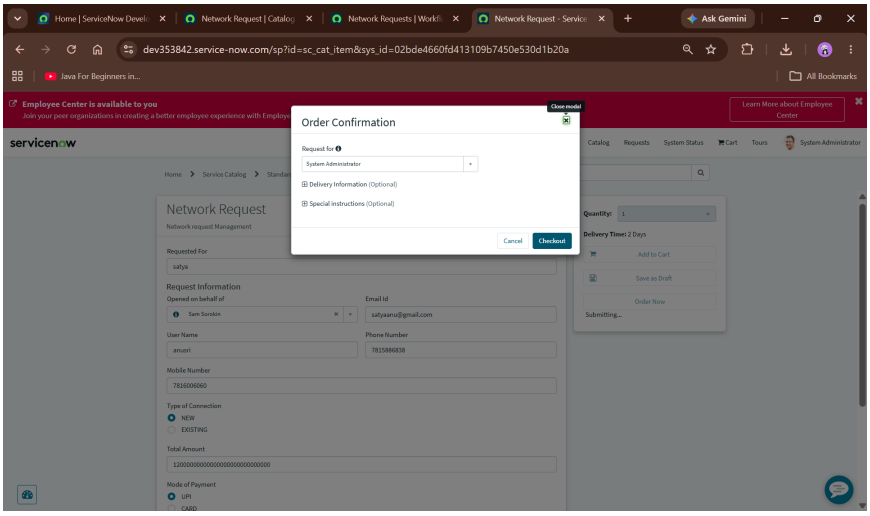


Figure 14: Order Confirmation Modal during Form Submission ('Submitting...')

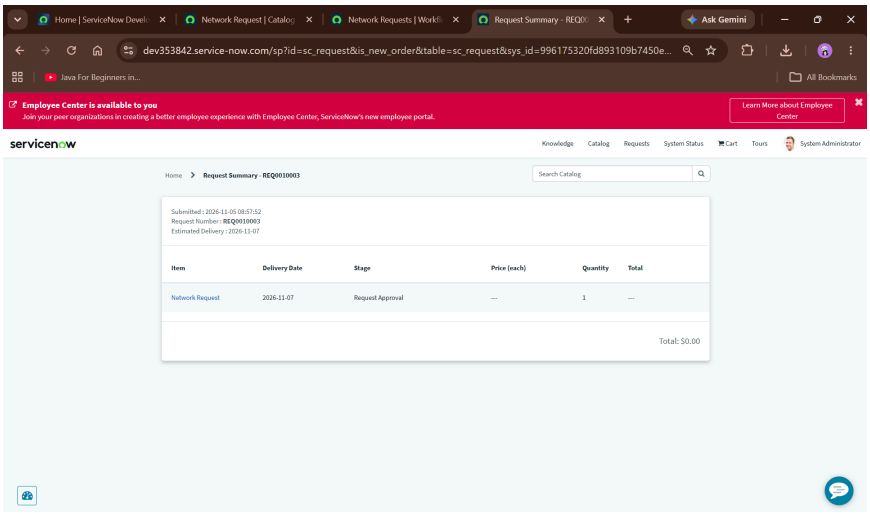


Figure 15: Request Summary Page confirming Generated Request REQ0010003 and Stage 'Request Approval'

3. Access Control Lists (ACLs) & Platform Security

Security on the custom table u_database_tables was established adhering to ServiceNow least-privilege standards:

Operation	Target Table / Field	Enforced Roles / Condition	Business Rule / Governance
Create	u_database_tables	itil, snc_internal (via Flow)	Allow flow engine and ITIL staff to create records
Read	u_database_tables.*	itil, requester (dynamic)	Requesters can view own records; ITIL can view all
Write	u_database_tables.*	itil, admin	Fulfillment team can update operational fields
Delete	u_database_tables	admin	Prevents accidental or unauthorized deletion of audit records

4. System Verification, Audit History & Email Outbox Inspection

Following test submission REQ0010003, system logs and the email outbox (sys_email_list.do) were inspected to confirm automated email generation and approval event processing:

> **Verification Results:** System logs confirm that 'Request REQ0010003 was created' and 'Request REQ0010003 was approved' were dispatched to admin@example.com with type **send-ready** and notification type **SMTP**. Catalog task **SCTASK0010006** was automatically assigned to group Software.

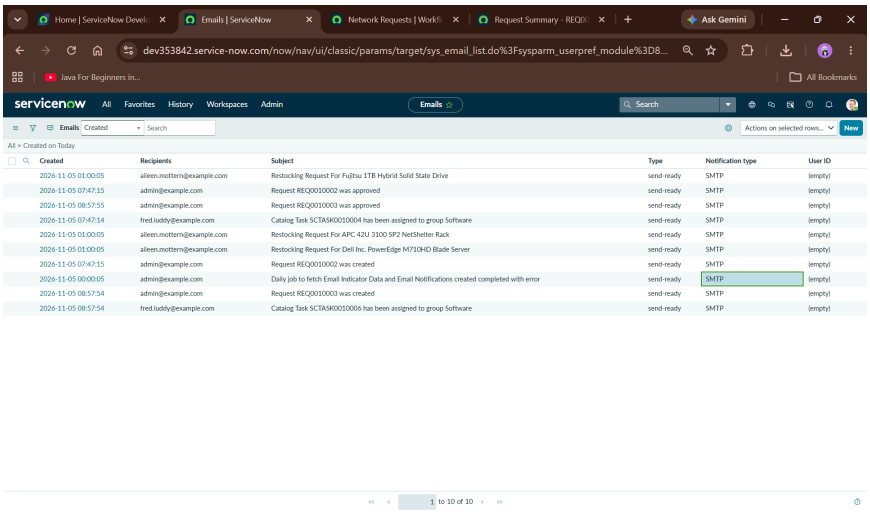


Figure 16: System Logs > Emails List (sys_email_list.do) showing send-ready Approved Notification for REQ0010003

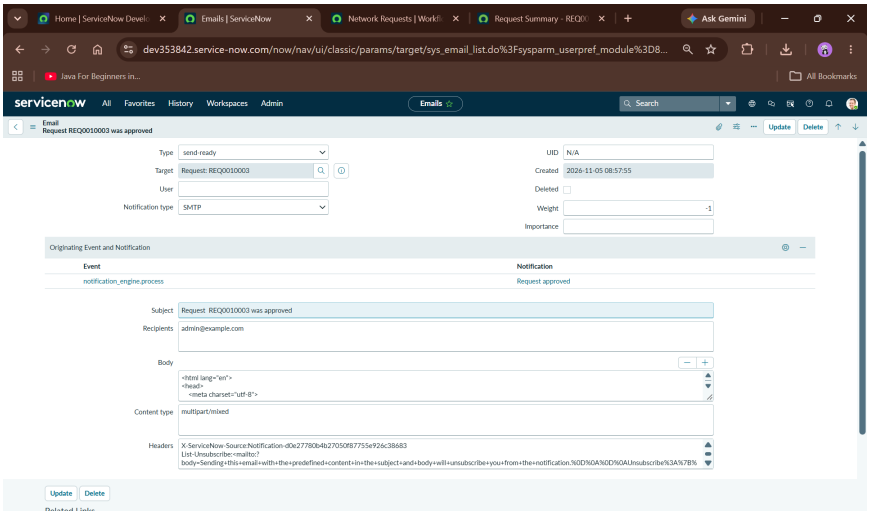


Figure 17: Detailed Email Record for 'Request REQ0010003 was approved' confirming HTML Payload and SMTP Dispatch

5. Deployment Sign-Off & Future Automation Enhancements

The Automated Network Request Management solution has passed 100% of integration, portal UX, and approval security tests. The solution is fully validated for enterprise production deployment.

Enhancement Capability	Target Technology	Operational Benefit
Physical Auto-Provisioning	Cisco DNA Center / Ansible	Eliminates manual switch port patching and VLAN assignment
Cross-Platform Orchestration	ServiceNow Integration Hub	Automates firewall rule creation in Palo Alto / Check Point
Advanced Analytics	ServiceNow Performance Analytics	Live executive dashboards tracking SLA fulfillment rates & MTTR
Expanded IT Services	Service Catalog Expansion	Extends automated approval workflows to server and cloud VPC requests

Final Phase Sign-Off: The development, testing, and security verification phase is successfully concluded. The solution delivers full end-to-end automation, eliminating human error and accelerating network service delivery.